

Show all of your work in the space provided to you. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

1. (20 pts) Consider the system of equations

$$\begin{array}{rcl} x + 5z & = & 2 \\ -2x + y - 6z & = & -1 \\ 2y + 8z & = & 6 \end{array}$$

- a) Write the system of equations as a **vector** equation.

- b) Write the system of equations as a **matrix** equation.

- c) Rewrite the augmented matrix in **echelon form (does not need to be reduced echelon form)**. **Show steps in the reduction.**

- d) What does this tell you about the **type** of solution for the system of equations?
(You do not need to solve the system.)

2. (12 pts) Let A be a 6×5 matrix. For each statement decide if it is always true, it is sometimes true and sometimes false, or it is always false. **No explanations required.**

- a) The columns of A span $\mathbb{R}^6$. Always true Always false Could be true or false

- b) The system $A\mathbf{x} = \mathbf{0}$ is consistent. Always true Always false Could be true or false

- c) The columns of A are linearly independent. Always true Always false Could be true or false

- d) The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ takes vectors in $\mathbb{R}^5$ and produces vectors in $\mathbb{R}^6$. Always true Always false Could be true or false

3. (20 pts) Given that the matrix A is row equivalent to

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

a) Write the solution of $A\mathbf{x} = \mathbf{0}$ in parametric vector form.

b) Describe the solution set geometrically.

c) Is the system $A\mathbf{x} = \mathbf{b}$ consistent for every $\mathbf{b}$ in $\mathbb{R}^4$? Explain.

d) Are there any vectors $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has a unique solution? Explain.

4. (9 pts) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation with $T(\mathbf{e}_1) = (1,2,3)$ and $T(\mathbf{e}_2) = (1,1,1)$

a) What is $T(3,10)$?

b) What is $T(x,y)$?

c) What is the standard matrix for T ?

5. (9 pts) Let $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 3 & 0 \end{bmatrix}$.

a) Write the second column of AB as a linear combination of the columns of A .

b) What is B^T ?

c) What is the 2,3 entry of $B^T A$?

6. (4 pts) What are the two properties that every linear transformation, T , must satisfy?

7. (8 pts) Let T be a linear transformation from $\mathbf{R}^5$ to $\mathbf{R}^3$ with standard matrix A .

a) What must be the size of the matrix?

b) Is it possible for the mapping, T , to be one-to-one? Explain your answer.

8. (6 pts) Let T be a linear transformation from $\mathbf{R}^5$ to $\mathbf{R}^3$ with standard matrix A . Circle each letter beside any statement listed below that is equivalent to the statement “The range of T is all of $\mathbf{R}^3$.”

a) There is a pivot in every row of $\text{RREF}(A)$.

b) There is a pivot in every column of $\text{RREF}(A)$.

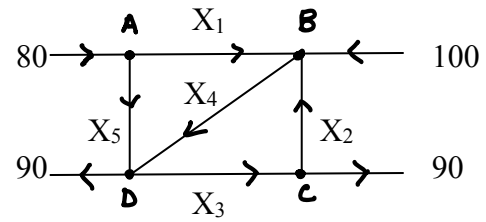
c) The matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent for every vector $\mathbf{b}$ in $\mathbf{R}^3$.

d) The columns of A are linearly independent.

e) $\mathbf{R}^3$ is equal to the span of the columns of A .

f) Every vector in $\mathbf{R}^3$ can be written as a linear combination of the columns of A .

9. (12 pts) Consider the network shown here.



a) Write a system of linear equations that models the network flow.

b) What is the **reduced echelon form** for the augmented matrix for this system?

c) If the flow along each edge must be nonnegative, find the maximum value possible for X_5 and the minimum value possible for X_4 ?